MIL-E-1/1328B
26 June 1972
SUPERSEDING
MIL-E-1/1328A
15 June 1961

MILITARY SPECIFICATION SHEET

ELECTRON TUBE, RECEIVING

TYPE 5842WA

The complete requirements for procuring the electron tube described herein shall consist of this document and the latest issue of Specification MIL-E-1.

This specification is approved for use by all Departments and Agencies of the Department of Defense.

DESCRIPTION: Miniature, High Gain, low noise, triode

Outline --- 6-1

Base --- E9-1

Envelope --- T6 1/2

Cathode --- Coated unipotential

Base connections:

Pin No.	1	2	3	4	5	6	7	8	9
Element	a	nc	h	g	g	k	g	g	h

ABSOLUTE RATINGS:

Parameter:	Ef	Eb	Ec	Ehk	Rk	Rg
Units:	V	Vdc	Vdc	V	ohms	Meg
Maximum:	6.9	200	---	100	----	0.5
Minimum:	5.7	---	---	---	----	---
Test conditions:	6.3	150	0	0	60	---

ABSOLUTE RATINGS:

Parameter:	Ik	Ic	Pp	T(envelope)	Alt
Units:	mAdc	mAdc	W	°C	ft.
Maximum:	38	3.0	4.5	±165	80,000
Minimum:	----	----	---	----	---
Test Conditions:	----	----	---	----	---

GENERAL:

Qualification - Required
Reliable

5842WA

FSC 5960

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METHOD	REQUIREMENT OR TEST	CONDITIONS	AQL (PERCENT DEFECTIVE)	INSPECTION LEVEL OR CODE	SYMBOL	LIMITS		UNIT
						Min	Max	
	Quality conformance inspection, part 1 (see notes 1 and 3)							
1266	Total grid current	Rg = 0.47 Meg, max	0.4	II	Ic	0	-0.5	μAdc
1256	Electrode current (anode) (1)		0.4	II	Ib	16	30	mAdc
1256	Electrode current (anode) (2)	Ec = -5 Vdc	0.4	II	Ib	---	100	μAdc
1301	Heater current		0.4	II	If	280	320	mA
1336	Heater cathode leakage	Ehk = +100 Vdc Ehk = -100 Vdc	0.4	II	Ihk Ihk	--- ---	7 7	μAdc μAdc
1306	Transconductance (1)	Ck = 1000 μF See note 2	0.4	II	Sm	20000	30000	μmhos
1201	Short and discontinuity detection		0.4	II	---	---	---	---
	Quality conformance inspection, part 2 (see note 3)							
1211	Insulation of electrodes	Eg - all = -100 Vdc Ep - all = -300 Vdc Ehk - all= -100 Vdc	2.5	S3	R R R	500 500 500	--- --- ---	Meg Meg Meg
1306	Transconductance (2)	Ef = 5.7 V Ck = 1000 μF	2.5	I	Sm	---	12	%
1266	Grid current	Ef = 7.5 V Ec = -10 Vdc Rk = 60 ohms Rg = 0.5 Meg See note 4	2.5	I	Isg	0	-0.5	μAdc
2201	Noise and microphonics	Ef = 6.3 Vac Ehk = 0, Ebb = 200 Vdc, Ec = 0 Ecal = 200 mVac Rp = 10,000 Rg = 0, Rk = 60 Ck = 1000 μF min	2.5	I	---	---	---	---
1316	Amplification factor	----------	2.5	S3	Mu	35	55	---
1351	Direct-interelectrode capacitance	No shield	2.5	Code E	Ck-G+H Cp-G+H Cp-K+H	8.0 1.5 ---	10 1.95 .55	pF pF pF
1002	Barometric pressure, reduced	Pressure = 21 ±2 mmHg; voltage = 500 Vac See note 5	6.5	S3	---	---	---	---
1031	Vibration	Ec = -2 Vdc Rp = 2000; G = 10; F = 40 Hz; Rk = 0	6.5	Code H	Ep	---	150	mVac
2126	Glass strain		2.5	I	---	---	---	---
1121	Base strain		---	---	---	---	---	---

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METHOD	REQUIREMENT OR TEST	CONDITIONS	AQL (Percent Defective)	Insp Level or Code	Allowable Defectives Per Characteristic		Sym	LIMITS		Unit
					1st sample	Combined samples		Min	Max	
	Quality conformance inspection, part 2 -continued									
1521	Survival rate life	Stability life conditions or equivalent	----	--	---	---	---	---	---	---
	Life test end points: (100 hours)									
		Continuity and shorts	0.65	II	---	---	---	---	---	---
		(inoperative transconductance (1)	1.0	I	---	---	Sm	18000	---	μmhos
1501	Intermittent life	Stability life conditions T envelope = +165°C Min See note 7	---	--	---	---	---	---	---	---
	Life test end points: (500 hours)									
		Inoperatives	---	--	1	3	---	---	---	---
		Grid current	---	--	1	3	Ig	0	-0.5	μAdc
		Heater current	---	--	1	3	If	280	320	mA
		Change in trans-conductance(1) of individual tubes	---	--	1	3	ΔSm_t	---	20	%
		Transconductance (2)	---	--	1	3	ΔSm_{Ef}	---	15	%
		Heater cathode leakage								
		Ehk = +100 Vdc	---	--	1	3	Ihk	---	10	μAdc
		Ehk = -100 Vdc					Ihk	---	10	μAdc
		Insulation of electrodes								
		g - all	---	--	1	3	R	100	---	Meg
		p - all	---	--			R	200	---	Meg
		Transconductance (1)	---	---			ΔSm_{avg}	---	15	%
		Total defectives	---	---	3	6	---	---	---	---
	Life test end points:(1000 hours)									
		Inoperatives	---	--	1	3	---	---	---	---
		Grid current	---	--	1	3	Ig	0	-0.5	μAdc
		Heater current	---	--	1	3	If	280	320	mA
		Change in trans-conductance(1) of individual tubes	---	--	1	3	ΔSm_t	---	25	%
		Transconductance (2)	---	--	1	3	ΔSm_{Ef}	---	20	%
		Heater cathode leakage								
		Ehk = +100 Vdc	---	--	1	3	Ihk	---	10	μAdc
		Ehk = -100 Vdc					Ihk	---	10	μAdc
		Insulation of electrodes								
		g - all	---	--	1	3	p	50	---	Meg
		p - all					p	100	---	Meg
		Total defectives	---	--	4	5	---	---	---	---

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METHOD	REQUIREMENT OR TEST	CONDITIONS	AQL (PERCENT DEFECTIVE)	INSPECTION LEVEL OR CODE	SYMBOL	LIMITS		UNIT
						Min	Max	
	Quality conformance inspection, part 3 See note 1							
1041	Shock	Hammer angle = 30° Ehk = +100 Vdc Ck = 0 See note 6	---	---	---	---	---	---
1031	Vibration	G = 2.5; fixed frequency F = 25 Hz min, 60 Hz max See note 5	---	---	---	---	---	---
----	Shock and vibration end points:							
1031	Vibration		---	---	Ep	---	300	mVac
1336	Heater-cathode leakage	Ehk = +100 Vdc	---	---	Ihk	---	20	μAdc
		Ehk = -100 Vdc	---	---	Ihk	---	20	μAdc
1306	Transconductance(1)		---	---	Sm	18000	---	μmhos
1266	Grid current		---	---	Ig	0	-1.0	μAdc
1506	Heater-cycling life	Ef = 7.5 V Ehk = +100 Vdc Rk = 0; Ec = 0; Eb = 0	---	---	---	---	---	---
	Life test end points:							
1336	Heater-cathode leakage	Ehk = +100 Vdc	---	---	Ihk	---	15	μAdc
		Ehk = -100 Vdc						
1516	Stability life	Ehk = +100 Vdc Rg = 0.47 Meg TA = Room	1.0	I	---	---	---	---
	Life test end points: (2 and 20 hours)							
1306	Transconductance(1)		---	---	ΔSmt	---	10	%

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NOTES:

1. Variables sampling procedure: See 4.3.1

2. The reactance of the cathode by-pass capacitor shall be 0.5 ohms maximum at the test frequency.

3. The AQL for the combined defectives for attributes in quality conformance tests, part 1, excluding inoperatives and mechanical, shall be 1 percent. When lot sizes of 500 tubes or less are submitted for acceptance, the 0.40 percent AQL for individual characteristics shall not apply.

4. Prior to this test, tubes shall be preheated a minimum of five minutes with all sections operating at the conditions indicated below. Three minute test is not permitted. Test at specified conditions within three seconds after preheating. Grid emission shall be the last test performed on the sample selected for the grid emission test.

Ef V	Ec Vdc	Eb Vdc	Rk ohms	Rg Meg
7.5	0	150	60	0.5

5. This test shall be conducted on the initial lot and thereafter on a lot approximately every 30 days. When one lot has passed, the 30-day rule shall apply. In the event of lot failure, the lot shall be rejected and the succeeding lots shall be subjected to this test until a lot passes. Standard, MIL-STD-105, sample size code letter F shall apply.

6. A grid resistor of 0.1 megohm shall be added; however, this resistor will not be used when thyratron-type short indicator is employed.

7. Envelope temperature is defined as the highest temperature indicated when using a thermocouple of no. 40 B & S or smaller diameter elements welded to a ring of 0.025-inch diameter phosphor bronze in contact with the envelope. Envelope temperature requirement will be satisfied if a tube, under normal test conditions, is determined to operate at minimum specified temperature at any position on the life test rack.

Custodians:
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Air Force - 80

Review activities:
Air Force - 11, 17

Preparing activity:
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